

Computer Organization and Architecture (24B41EC312)



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Module 1:

STRUCTURE OF COMPUTERS:

- Generation of Computers,
- Computer types,
- Classification of Computers, functional units, basic operational concepts,
- Von-Neumann architecture, bus structures, types of bus, software,
- Multiprocessors and multicomputer.

Von-Neumann architecture

- **Definition:** Von Neumann architecture is a computer design where instructions and data are stored in the same memory space. This means the CPU fetches both instructions and data from the same memory, using the same pathways.
- The **Von Neumann Architecture** is the basic blueprint for nearly every computer you use today, from your smartphone to your laptop. Proposed by John von Neumann in 1945, it changed computers from fixed machines (like a calculator that only does math) into **programmable** machines that can do anything based on the software they run.
- 1. The Core Idea: One "Locker" for Everything

Before this design, computers kept their instructions (the "to-do list") and their data (the "materials") in separate places. Von Neumann's big idea was to store both instructions and data in the same memory.

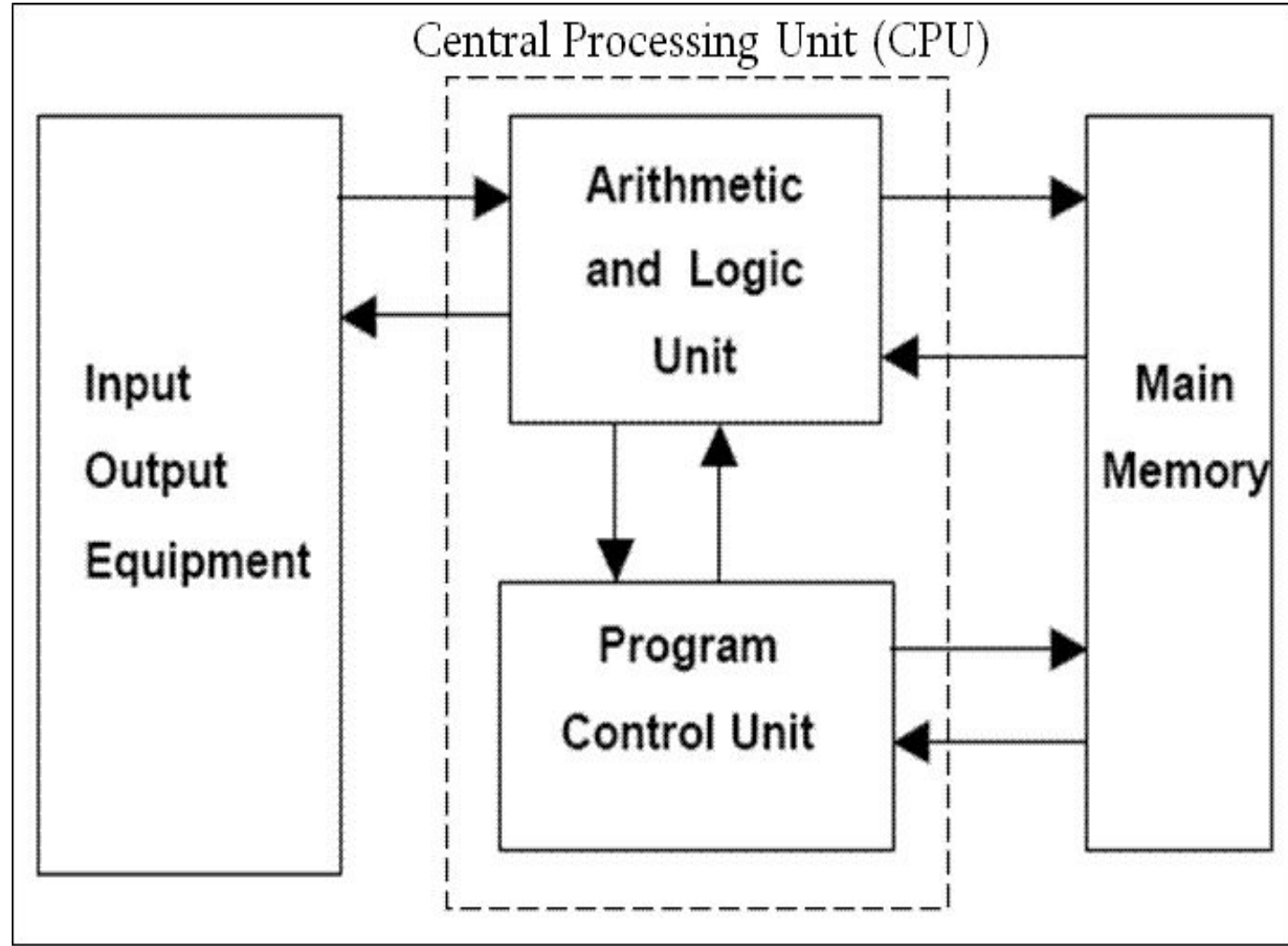


Figure : General structure of Von Neumann Architecture

- Von-Neumann architecture consists of The Four Main Parts:

1. **Central Processing Unit (CPU):** The "Brain." It does all the thinking and carries out commands.
 - a. **ALU (Arithmetic Logic Unit):** The calculator part that does math (adding, subtracting) and logic (deciding if one thing is bigger than another).
 - b. **Control Unit:** The manager that directs traffic and tells all other parts what to do.
2. **Memory (RAM):** The "Workspace." A temporary storage area where the CPU keeps the program it's currently running and the data it's using.
3. **Input/output (I/O) Devices:** How the computer talks to us. Keyboards and mice are Inputs; screens and printers are Outputs.
4. **Bus:** The "Highway." A set of wires that carries information back and forth between the CPU and memory.

How It Works (The Cycle)

1. The computer follows a never-ending loop called the Fetch-Decode-Execute Cycle:
2. Fetch: The CPU grabs the next instruction from memory.
3. Decode: The manager (Control Unit) figures out what the instruction means (e.g., "Add these two numbers").
4. Execute: The CPU does the actual work.
5. Store: It saves the result back into memory and starts over.

The One Big Problem: The "Bottleneck"

- Because instructions and data share the same "highway" (the bus), the CPU often has to wait. It can only grab **one thing at a time**—either a piece of data or an instruction. This is known as the **Von Neumann Bottleneck**,
- it is the main reason your computer can sometimes feel slow even if the processor is powerful

Applications

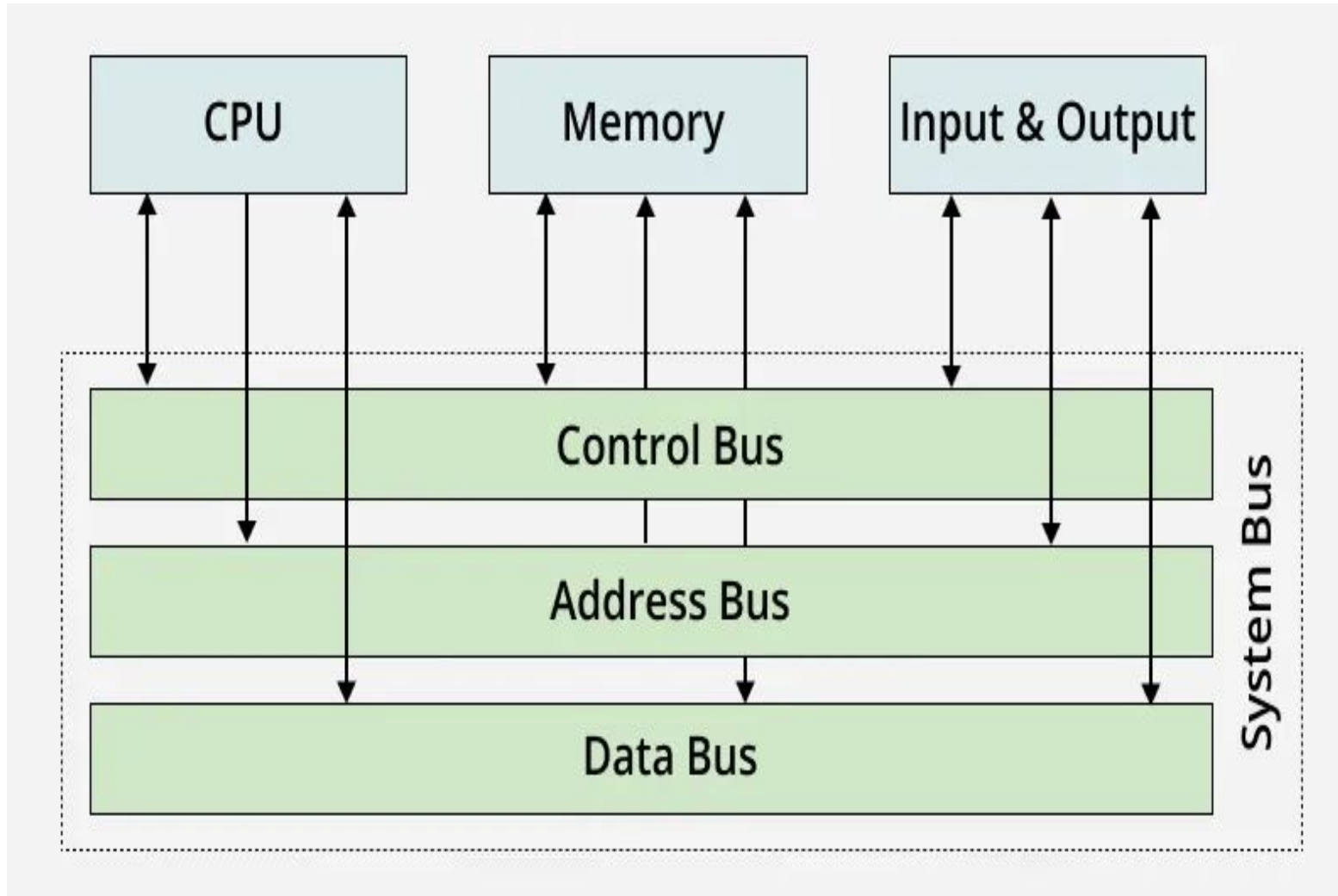
- Von Neumann architecture is the foundation of most modern computing systems, where both instructions and data are stored in the same memory. Here are the some applications:
- **Personal Computers and Laptops:** Most PCs and laptops (Intel/AMD) use a shared memory that stores both programs and data, enabling efficient running of software like Windows and browsers. **Example:** Intel Core i7, AMD Ryzen processors.
- **Smartphones and Tablets:** ARM-based devices use the shared memory model to support multitasking and power efficiency in apps on iOS and Android. **Example:** Qualcomm Snapdragon, Apple A-series chips
- **Embedded Systems:** Microcontrollers in cars, IoT devices, and appliances use Von Neumann architecture for simple, cost-effective processing of instructions and data in the same memory. **Example** Arduino Uno (ATmega328), older ARM Cortex-M designs.
- **Servers and Cloud Computing:** Servers utilize shared memory systems to handle large-scale software and data tasks for services like AWS and Netflix.
- **Gaming Consoles:** Consoles like PS5 and Xbox use unified memory to run game code and manage inputs, enabling smooth gameplay.

BUS STRUCTURES:

- A computer bus is a communication system used to transfer data between components within a computer or between different computers. It plays an important role in minimizing the number of connections needed by centralizing communication over shared pathways.
- It consists of physical connections like wires, circuits, or cables.
- Components like the CPU, memory, and input/output (I/O) devices are connected through a bus.
- It simplifies data transfer and improves efficiency.

Types Of Buses

There are three main types of buses in a computer system, which are discussed below:



1. Address Bus

- A collection of wires used to identify particular location in main memory is called Address Bus. Or in other words, the information used to describe the memory locations travels along the address bus.
- The address bus transports memory addresses which the processor wants to access in order to read or write data..
- The address bus is unidirectional.
- The size of address bus determines how many unique memory locations can be addressed.

Example:

- A system with 4-bit address bus can address $2^4 = 16$ Bytes of memory.
- A system with 16-bit address bus can address $2^{16} = 64$ KB of memory
- A system with 20-bit address bus can address $2^{20} = 1$ MB of memory.

2. Data Bus

- A collection of wires through which data is transmitted from one part of a computer to another is called Data Bus. It can be thought of as a highway on which data travels within a computer.
- The main objective of data bus is transfer of the data between microprocessor to input/ output devices or memory.
- The data bus transfers instructions coming from or going to the processor.
- The data bus is bidirectional because the data can flow in either direction from CPU to memory(or input/output device) or from memory to the CPU.
- The size (width) of bus determines how much data can be transmitted at one time.
- Example:
 - A 16-bit bus can transmit 16 bits of data at a time.
 - 32-bit bus can transmit 32 bits at a time.

3. Control Bus

- The connections that carry control information between the CPU and other devices within the computer is called Control Bus. The control bus transports orders and synchronization signal coming from the control unit and travelling to all other hardware components
- The main objective of control bus is all signals controller carried from processor to other hardware device.
- The Control bus is bidirectional because the data can flow in either direction from CPU to memory(or input/output device) or from memory to the CPU.
- It also transmits response signals from the hardware.

Comparison Between System Buses

- The below table shows the comparison between the three buses as below :

Buses	Purpose & Key Role
Address Bus (Unidirectional)	Carries memory addresses; Identifies where data should go
Data Bus (Bidirectional)	Carries actual data, moves data between components
Control Bus(Bidirectional)	Carries control and sync signals, coordinates CPU and device actions

SOFTWARE

- Software/Computer Software refers to the collection of instructions, data, or programs that are used to operate computers and execute specific tasks. These are important for the functioning of modern computing systems.

Key Features:

- **Intangible:** Computer software cannot be seen or touched like hardware.
- **Provides User Interface:** Computer Software provides a user interface that allows users to interact with the computer.
- **Enhances Efficiency:** Efficient computer software makes optimal use of system resources such as memory, processing power, and storage.
- **Portability:** Computer software is designed with the ability to run on different hardware platforms or operating systems with minimal modifications.
- **Documentation:** Detailed comprehensive documentation provides information on how to install, use, and maintain the computer software.

Types of Computer Software

Here are the different types of Computer Software based on their functionalities, purposes, and usage:

1. System Software

- System software manages hardware and basic system operations and serves as a foundation for application software. It serves as a bridge between hardware and user and ensures that the system operates efficiently and effectively. This includes Operating Systems, Device Drivers, and Utility Software.

2. Application Software

- Application Software refers to the programs designed to perform specific tasks for end-users such as web browsing, gaming, word processing, etc. This includes Productivity Software (Microsoft Office, Google Workspace), Database Software (MySQL, Oracle Database), Communication Software (Slack, Skype), and Graphics Software (Adobe Photoshop, Coral Draw).

3. Development Software

- Development Software includes tools and environments that are used by developers and programmers to create, debug, and maintain software applications. This includes Integrate Development Environments (Eclipse, Microsoft Visual Studio), Code Editors (Atom, Visual Studio Code), and Version Control Software (Git, GitHub).

4. Business Software

- Business Software refers to the applications that assist organizations in managing their operations, improving productivity, and facilitating various business activities. This includes Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), and Accounting Software (QuickBooks, Xero).

5. Embedded Software

- Embedded software is computer software that is designed to operate hardware and perform specific tasks within a larger system. It is optimized for particular functions and integrated into the hardware it controls. This includes Firmware (BIOS in computer) and Control Systems (Software in automotive control systems)

How is Computer Software Distributed?

- Here are some of the primary ways in which software is distributed:
- **Physical Media:** This includes CDs/ DVDs and USB Drives. Software is packaged and distributed on these physical media, often with installation instructions.
- **Digital Downloads:** Users download the software directly from the developer's website or platforms like Apple App Store, Google Play Store, and Microsoft Store.
- **Cloud-based Services:** Users access the software over the internet on a subscription basis or some software can be stored and accessed via cloud storage services like Dropbox or Google Drive.
- **Open-source Repositories:** Developers can share and distribute open-source software projects through platforms like GitHub.
- **Pre-installed on Devices:** Software is often pre-installed on computers, smartphones, and other devices before they are sold to customers.

How Does Computer Software Work?

Here are the explanation of how the computer software works:

- **Coding:** Software developers write the instructions using programming languages like Java, Python, etc. These instructions are in the form of algorithms and logic to achieve specific tasks.
- **Compilation/ Interpretation:** Depending upon the programming language the code is either compiled or interpreted. Programming languages like C, C++, and C# in which code is compiled and programming languages like PHP, Python, and Ruby in which code is interpreted.
- **Execution:** Once the software is compiled/ interpreted, it can be executed by the CPU. The CPU reads and executes the machine instructions generated from the code.
- **Interaction:** The software interacts with OS and hardware to perform tasks as instructed.
- **Output:** The software produces output based on its instructions which may include generating reports, controlling hardware devices, etc.

Computer Software Examples

Here are some of the examples of Computer Software:

- **Operating Systems:** These include Windows, macOS, Linux, and Mobile OS such as Android, and iOS.
- **Productivity Software:** This includes Microsoft Office, Google Workspace, Note-taking software such as Evernote, and OneNote, and Project Management Software such as Trello, and Microsoft Project.
- **Web Browsers:** Some of the popular web browsers are Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.
- **Security Software:** Security software includes Antivirus, Firewall, and Password Managers.
- **Database Management Software:** DBMS includes Relational Database such as MySQL, PostgreSQL, and Oracle Database and Non-Relational Database such as MongoDB, and Redis.

Future of Computer Software

Here are some of the developments expected to affect the future of software:

- 1. Artificial Intelligence and Machine Learning:** AI and ML will enhance the capabilities of the software such as automation, data analysis, and personalized user experiences.
- 2. Quantum Computing:** Software will need to be developed to use quantum computing to solve complex problems faster than the classic computers.
- 3. Internet of Things (IoT):** Software will need to manage, analyze, and secure the vast amount of data generated by IoT devices.
- 4. Augmented Reality (AR) and Virtual Reality (VR):** Software will be more focused on creating realistic interactive environments using AR and VR.
- 5. Decentralized Applications (DApps):** DApps will gain popularity in finance, supply chain, and voting systems.

Introduction of Multiprocessor and Multicomputer

- Think of a Multiprocessor (Shared Memory) as one big brain (a single computer) with several thinking parts (CPUs) that share the same desk (memory) and whiteboard (data), working closely together on one big task. A Multicomputer (distributed memory), on the other hand, is a team of separate, independent computers (like PCs) linked by a network, each with its own desk and whiteboard, communicating by passing notes (messages) to solve a problem together.

- In layman's terms, both multiprocessors and multicomputer are ways to make computing faster by using "more hands," but they differ in how those hands share their workspace.

1. Multiprocessor (The "One Kitchen" Model)

- Think of a Multiprocessor like a single, large professional kitchen.
- How it works: There are several chefs (CPUs/Processors) all working inside the same room.
- Sharing: They all share the same pantry, fridge, and stove (Shared Memory/RAM).

- Communication: If one chef needs to tell another the soup is ready, they just leave it on the counter. Communication is "implicit" because they are all looking at the same stuff.
- Pros/Cons: It is very fast and easy to coordinate, but if the fridge breaks, everyone is stuck. Also, if you add too many chefs, they start bumping into each other in the small space (limited scalability)

2. Multicomputer (The "Restaurant Chain" Model)

- A Multicomputer is like a chain of independent restaurants (like several McDonald's locations) working on a massive order.
- How it works: Each location is its own standalone building (Autonomous Computer) with its own chef, its own pantry, and its own fridge.
- Sharing: They do not share a pantry. If one restaurant runs out of salt, it can't just reach into another's cupboard.

Communication: To work together, they have to send messages to each other (e.g., via phone or delivery truck) over a network. This is called "Message Passing".

- Pros/Cons: It is very easy to grow—you just build more restaurants (high scalability). If one location burns down, the others keep cooking. However, it takes longer to coordinate because they have to wait for messages to travel between buildings

Key difference between multiprocessor & multicomputer

- The fundamental difference between a multiprocessor and a multicomputer lies in **memory sharing and coupling**: a multiprocessor is a single computer with multiple CPUs sharing a common, central memory, whereas a multicomputer consists of multiple independent computers, each with its own private memory, connected via a network.

Difference b/w Multiprocessor & Multicomputer

Feature 	Multiprocessor	Multicomputer
System Type	A single computer with many "brains".	A group of independent computers.
Memory	Shared: Everyone uses the same RAM.	Private: Each has its own RAM.
Connection	Tightly coupled (physically close).	Loosely coupled (connected by network).
Ease of Use	Easier to program; they "see" the same data.	Harder to program; must coordinate messages.
Reliability	If the shared memory fails, the whole system fails.	If one computer fails, others continue.

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